

Prometheus

This is an open source tool for monitoring and alerting. The Prometheus ecosystem has components such as a Prometheus server, client libraries, a push gateway, exporters for services, an alert manager, and support tools.

Prometheus has features such as a multi-dimensional data model, a flexible query language, and multiple modes of graphing and dashboarding support.

Initial release	2012
Stable release	v2.32.0
Written in	Go
Licence	Apache License 2.0
Website	https://prometheus.io/
GitHub repository	https://github.com/prometheus/prometheus
Features	- A multi-dimensional data model - A flexible query language (PromQL) - Support for service discovery or static configuration
How is it useful in implementing DevOps practices?	Monitoring
Is a commercial flavour available?	Prometheus is 100 per cent open source and community-driven

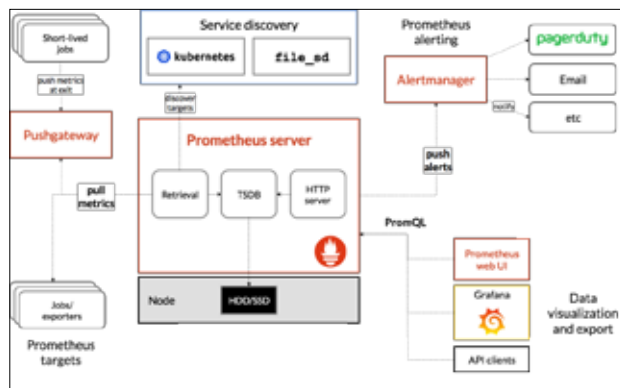


Figure 5: Prometheus architecture

(Source: <https://prometheus.io/docs/introduction/overview/>)

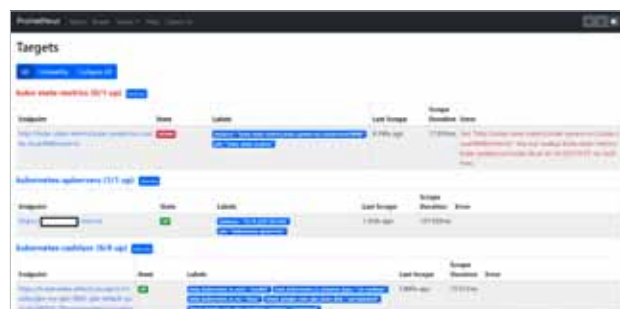


Figure 6: Prometheus dashboard

Terraform

This open source 'Infrastructure as Code' tool, developed by HashiCorp, helps in creating immutable infrastructure to avoid the problem of configuration drift. Terraform is a platform-agnostic declarative coding tool that helps developers to use a high-level configuration language to manage infrastructure. It supports different cloud providers like AWS, Azure, Google Cloud Platform, and so on.

Initial release	2014														
Stable release	1.0.10														
Written in	Go														
Licence	Mozilla Public License v2.0														
Website	https://www.terraform.io/														
GitHub repository	https://github.com/hashicorp/terraform														
Features	- Manages Infrastructure as Code using declarative config files - Dependency graphs - State management														
How is it useful in implementing DevOps practices?	Infrastructure as Code														
Can we integrate it with Pipeline as Code?	Yes. It can be part of Pipeline as Code, and different strategies can be used to create resources and deploy applications in different cloud platforms.														
Important Terraform commands	<table border="1"> <thead> <tr> <th>Description</th><th>Command</th></tr> </thead> <tbody> <tr> <td>Verify Terraform version</td><td><code>terraform --version</code></td></tr> <tr> <td>Install Terraform modules</td><td><code>terraform init</code> <code>terraform init -input=false</code></td></tr> <tr> <td>Check configuration and create a plan</td><td><code>terraform plan</code></td></tr> <tr> <td>Execute the plan</td><td><code>terraform apply</code> <code>terraform apply -auto-approve</code></td></tr> <tr> <td>Delete all resources</td><td><code>terraform destroy</code> <code>terraform plan -destroy</code></td></tr> <tr> <td>Validate/check the syntax of the Terraform files</td><td><code>terraform validate</code></td></tr> </tbody> </table>	Description	Command	Verify Terraform version	<code>terraform --version</code>	Install Terraform modules	<code>terraform init</code> <code>terraform init -input=false</code>	Check configuration and create a plan	<code>terraform plan</code>	Execute the plan	<code>terraform apply</code> <code>terraform apply -auto-approve</code>	Delete all resources	<code>terraform destroy</code> <code>terraform plan -destroy</code>	Validate/check the syntax of the Terraform files	<code>terraform validate</code>
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Tutorials	https://learn.hashicorp.com/terraform														

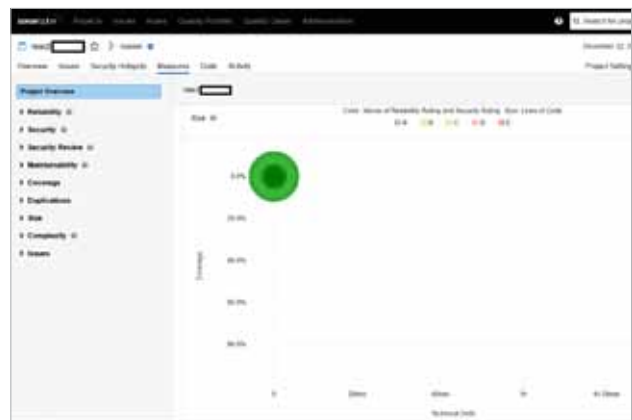


Figure 7: SonarQube dashboard